

MAYANK SHARMA

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AI SOFTWARE ENGINEERING PROFILE

Computer Science Engineer specializing in **Generative AI, Large Language Models (LLMs), Backend Engineering, and Scalable Software Systems**. Experienced in building LLM-powered applications, asynchronous backend workflows, REST APIs, secure services, and database-driven systems using **Node.js, Express.js, MongoDB, React.js, and OpenRouter**. Strong interest in developing reliable AI solutions with emphasis on **scalability, fault tolerance, security, system design, and production-oriented engineering**.

PROJECTS

AI Customer Support Deflector – LLM-Powered Email Intelligence Platform

Tech: Node.js, Express.js, JavaScript, REST APIs, OpenRouter API, LLMs, Prompt Engineering, NLP

- Engineered a **Generative AI-powered email intelligence platform** that analyzes customer emails, classifies intent, detects sentiment, identifies issue categories, and generates context-aware response drafts using LLMs through the OpenRouter API.
- Designed modular AI inference workflows for **intent classification, sentiment analysis, issue categorization, contextual response generation, and prompt optimization**.
- Implemented reusable **prompt engineering strategies, asynchronous API workflows, response validation, fallback handling, and output refinement** to improve reliability and consistency of model-generated responses.
- Integrated Node.js backend services with REST APIs and LLM inference pipelines to provide real-time AI-assisted customer support recommendations.

File Processing System – Distributed Asynchronous Processing Platform

Tech: Node.js, Express.js, MongoDB, Mongoose, REST APIs, Asynchronous Processing, Worker-Based Architecture

- Designing a **distributed file-processing system** for handling large batches of CSV files through asynchronous job processing rather than performing resource-intensive work directly inside API requests.
- Designed **Batch, File, Job, and Worker state models** to track uploaded files, processing status, results, retries, failures, and ownership throughout the processing lifecycle.
- Implemented the architecture around **worker-based job execution**, where workers claim queued jobs and persist their worker ID, processing status, retry count, and lease/heartbeat information in MongoDB.
- Designed for **fault tolerance and scalability** using job retries, persistent processing state, checkpointing, partial-result persistence, and horizontal worker scaling to prevent a single worker failure from losing an entire batch.

SmartMoney – Full-Stack Personal Finance Platform

Tech: React.js, JavaScript, Node.js, Express.js, MongoDB, JWT, bcrypt, REST APIs, Docker, Docker Compose, Vite

- Developed a secure end-to-end personal finance platform integrating **expense tracking, budgeting, savings goals, investments, recurring payments, analytics, and financial reporting**.
- Built **Node.js/Express REST APIs**, MongoDB-backed data models, reusable backend services, protected routes, and responsive React interfaces.
- Implemented authentication and backend security using **JWT, bcrypt, protected routes, Helmet, CORS, and rate limiting**, with Docker-based development and deployment workflows.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), Python, C++, HTML5, CSS3

Artificial Intelligence: Generative AI, Large Language Models (LLMs), Prompt Engineering, OpenRouter API, NLP, Sentiment Analysis, Intent Classification, Response Generation, AI Integration, AI Workflow Design

Backend & Distributed Systems: Node.js, Express.js, REST APIs, Middleware, Asynchronous Processing, Worker-Based Systems, Queueing Concepts, Job Tracking, Retry & Recovery, Fault Tolerance, System Design

Databases: MongoDB, Mongoose, Data Modeling, CRUD, Database Design, Persistent Job State

Frontend: React.js, React Router, Vite, Responsive Design, Component Architecture, Framer Motion

Security & Architecture: JWT Authentication, bcrypt, Protected Routes, Helmet, Rate Limiting, CORS, REST Architecture, Full-Stack Architecture, API Security

Developer Tools: Git, GitHub, Docker, Docker Compose, Postman, VS Code, Chrome DevTools, NPM, Vercel

Core CS: Data Structures & Algorithms, Object-Oriented Programming, Asynchronous JavaScript, Promises, Event Loop, Software Development Lifecycle

EDUCATION

B.E. Computer Science Engineering (Artificial Intelligence & Machine Learning) – Chandigarh University

2020 – 2024

CERTIFICATIONS

- NLP with Attention Models
- NLP with Probabilistic Models
- NLP with Sequence Models
- MLOps Tools: MLflow & Hugging Face

ACHIEVEMENTS

Cleared **AFCAT (Air Force Common Admission Test)** – 3 Times | Captain of **Zonal & District Football Teams**, leading 20+ players